Arunabha Majumder

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NEURA Robotics, MiPA Robotics team

Application: Senior Mechanical Engineer, MiPA

Dear NEURA Robotics Hiring Team,

I am applying for the Senior Mechanical Engineer role in the MiPA Robotics team. Robots that work safely and close to people are what I have built in my PhD at Aalborg University, so MiPA is a team I would really like to join.

My main project was a hybrid shoulder exoskeleton. I designed the mechanism myself: a variable stiffness mechanism, pretensioned by a worm gear, working in parallel with a 6 Nm motor. I optimised its diameter, spring stiffness and pretension with a genetic algorithm, made the CAD models and drawings in SolidWorks, 3D printed prototypes and then had the parts manufactured. I integrated the motor, encoder and torque/force sensors and measured the stiffness on a physical test setup. It uses more than 25% less motor energy and won the Best Paper Award at IFToMM ISRM 2026. When a clutch part kept breaking and was hard to make, I replaced it with a simple bolt-locking clutch, a small lesson in Design for Manufacturing that stayed with me.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms, estimated its stiffness with less than 10% error and quantified human arm impedance. Across these projects I worked closely with control engineering and software colleagues, which is how I like to work.

I want to be honest: I do not have 10 years in industry, and I have not yet run series production, PDM/PLM systems or injection moulding projects. If this senior role is a step too far, I would be glad to be considered for a mechanical engineer position in the MiPA team. My research is finished, so I can start soon. Thank you for considering my application.

Kind regards,
Arunabha Majumder